

Faithfulness Under Pressure

Context and History Pruning for Conversational RAG Faithfulness

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Question

When a small language model receives long, noisy retrieved context, can pruning make its answer more faithful to the evidence?

- Task: conversational open-domain question answering with RAG.
- Main benchmark: QReCC, where each question depends on previous conversation turns.
- Output: short answer generated from retrieved/pruned evidence and retained dialogue history.
- Main risk: the model may keep stale history, miss the current evidence, or hallucinate unsupported claims.

Long context is not free.

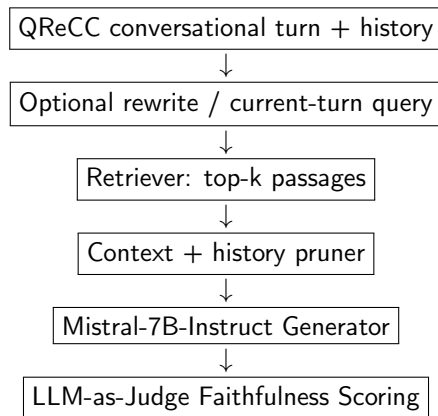
- More tokens increase latency and memory cost.
- Retrieved passages and old dialogue turns often contain near-misses, stale entities, and distractors.
- Small LMs are especially sensitive to noisy evidence.

Source: Liu et al., TACL 2024, “Lost in the Middle”.¹

Long context is not always used well.

- Liu et al. show that models often perform best when relevant information is near the beginning or end of the context.
- Performance can drop when evidence is in the middle, even for long-context models.

¹<https://aclanthology.org/2024.tacl-1.9/>



- The retriever and generator are held fixed across pruning strategies.
- Only retrieved context and retained history change.
- This isolates the effect of pruning on faithfulness, answer quality, and compression.

Pruning Strategies

ID	Strategy	Role
B1	No pruning	Full retrieved context; maximum evidence and maximum noise.
B2	Naive truncation	Hard token budget; common but relevance-agnostic.
B3	RECOMP-style extractive	Selects query-relevant sentences from retrieved passages.
S1	LLMLingua-2	Token-level compression using a bidirectional encoder classifier.
S2	Provenance	Sentence-level sequence-labeling pruner, also related to reranking.
S3	History pruning	Removes redundant context using conversation history.
S4	Combined	Sentence pruning plus history-aware pruning.
S5	CfC/LNN extension	Exploratory adaptive pruning model for multi-turn context.

- **RECOMP**: compresses retrieved documents before generation; reports compression rates as low as 6% with minimal performance loss in evaluated settings.²
- **LLMLingua-2**: formulates prompt compression as token classification and reports $1.6\text{--}2.9\times$ end-to-end latency acceleration at $2\text{--}5\times$ compression.³
- **Provence**: treats context pruning as sequence labeling and reports negligible to no performance drop at almost no additional RAG pipeline cost.⁴

Our angle

We use QReCC because the central problem is conversational: deciding which past turns and retrieved passages still matter for the current question.

²https://proceedings.iclr.cc/paper_files/paper/2024/hash/bda88ed2892f5e61c9a9bf215c566913-Abstract-Conference.html

³<https://huggingface.co/papers/2403.12968>

⁴<https://arxiv.org/abs/2501.16214>

Main benchmark: QReCC

- 14K conversations.
- About 81K question-answer pairs.
- Built for question rewriting in conversational context.
- Retrieval collection: 10M web pages split into 54M passages.
- Each sample includes original question, context, rewrite, answer, and answer provenance.

Current experiment status

- Faithfulness: fraction of generated claims supported by context.
- EM/F1: lexical answer correctness against gold answer.
- Compression ratio: output tokens / input tokens.
- History retention: retained turns / original turns.
- Pilot HotpotQA runs remain only as implementation sanity checks.

Source: Anantha et al., NAACL 2021 / QReCC dataset release.⁵

⁵<https://aclanthology.org/2021.naacl-main.44/>

Pilot Results: HotpotQA Sanity Run

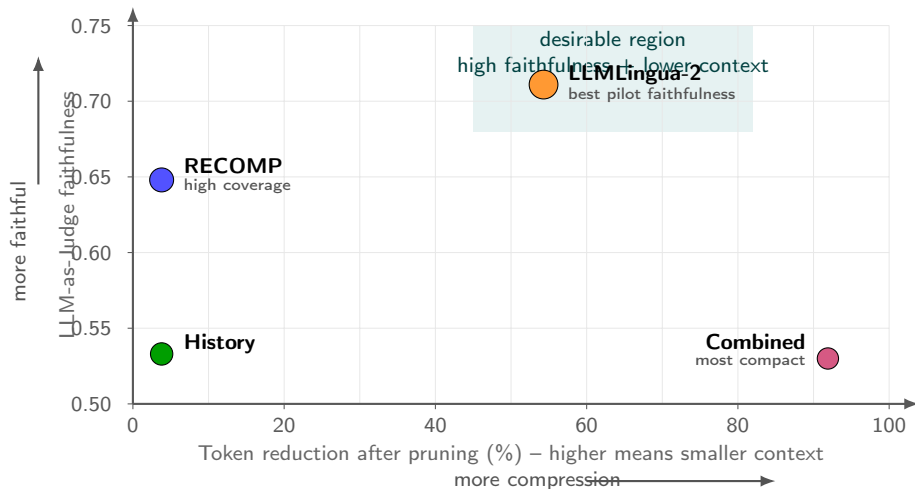
Strategy	Faith.	EM	F1	Comp.	Lat. s
B1: No pruning	0.718	0.040	0.148	1.000	6.76
B2: Naive truncation	0.621	0.040	0.139	0.918	4.95
B3: RECOMP	0.750	0.040	0.141	0.911	5.38
S1: LLMingua-2 fallback	0.750	0.040	0.141	0.911	5.38
S2: Provence	0.657	0.000	0.057	0.072	4.49
S3: History, empty	0.718	0.040	0.148	1.000	6.34
S4: Combined, empty	0.657	0.000	0.057	0.072	4.48

Notebook 2, 50 HotpotQA validation samples. These are pilot implementation checks, not the final QReCC benchmark results.

Strategy	Faith.	EM	F1	Comp.	Cov.	Noise
B3: RECOMP	0.648	0.080	0.195	0.962	0.760	0.624
S1: LLMingua-2 real	0.711	0.040	0.167	0.457	0.000	0.792
S3: History two-turn	0.533	0.100	0.214	0.962	0.627	0.641
S4: Combined two-turn	0.530	0.040	0.112	0.081	0.222	0.284

- LLMingua-2 is the strongest faithfulness result in this pilot run.
- Combined pruning is extremely compact and low-noise, but loses too much support.
- QReCC is the correct next benchmark for testing real history pruning.

Faithfulness–Compression Trade-off



Current numbers are HotpotQA pilot outputs; final benchmark target is QReCC conversational QA.

- **Pruning can improve faithfulness**, but not every pruning method helps equally.
- **Aggressive compression is dangerous for conversational QA**: if the relevant history turn disappears, the current question becomes ambiguous.
- **Faithfulness and correctness are different**: a faithful answer can still be the wrong answer, and a correct short answer can be judged poorly.
- **Exact coverage is too strict for token compression**: LLMLingua-2 may fragment evidence so gold sentences no longer match verbatim.
- **Conversation history is not just extra text**: old turns can help or distract depending on the current question.

Error Analysis Examples

Error type	Example pattern
Evidence removed	The “Arizona” question loses the sentence saying the song was written by Kenny Young.
Verbose correct answer	For yes/no questions, the model may answer correctly with explanation but get EM = 0.
Judge false negative	A short correct phrase such as “Political correctness” can be scored as unfaithful.
Entity near-miss	The generator outputs “T. M. Howard” instead of “T. R. M. Howard”.

Takeaway

We need judge calibration and qualitative analysis before treating one metric as ground truth.

Where the idea came from

When reading dense, information-rich text, humans slow down. High-entropy words take longer to process because they carry more unexpected information. Can we use this same signal to decide which conversation turns deserve more “processing time” in a neural network?

- Fixed thresholds do not know whether a turn is easy, noisy, redundant, or depends on earlier dialogue.
- We explore Closed-form Continuous-time networks (CfCs): temporal dynamics without a heavy ODE solver.⁶
- Key insight: map per-turn *surprisal* to a continuous time step Δt_i , so surprising turns get a wider integration window in the network's ODE.

⁶<https://www.nature.com/articles/s42256-022-00556-7>

Surprisal theory (psycholinguistics)

Reading time on a word is proportional to its surprisal $-\log P(w_j \mid w_{<j})$. A word that violates expectations takes more cognitive effort to integrate.

- Levy (2008): syntactic surprisal predicts word-by-word reading times.
- Smith & Levy (2013): the relationship is *logarithmic* and holds across many corpora.
- **Transfer to NLP:** if a turn is lexically surprising (high surprisal from a proxy LM), it likely introduces new information — and is more likely to be important for answering the current question.
- We do not claim surprisal *equals* importance; we ask whether it is a *useful proxy* — and test this with a simulation before training.

Levy 2008, Cognition 106(3). Smith & Levy 2013, Cognition 128(3).

From Reading Time to the CfC Time Step

For utterance u_i , compute surprisal with DistilGPT-2 (proxy LM):

$$S(u_i) = -\frac{1}{N} \sum_{j=1}^N \log P(w_j \mid w_{<j})$$

Map surprisal to a continuous time step (rank-normalised for outlier robustness):

$$\Delta t_i = \Delta t_{\min} + \beta \cdot \frac{\text{rank}(S_i)}{T}$$

The CfC network then integrates over Δt_i at each turn. The *commitment fraction* measures how strongly the state “locks on”:

$$\varphi_i = 1 - e^{-\lambda \Delta t_i}$$

- Large $\Delta t_i \rightarrow$ large $\varphi_i \rightarrow$ turn leaves a stronger trace in the hidden state $\rightarrow$ higher importance score.
- Rank normalisation prevents one outlier surprisal from dominating the time budget (validated by Simulation S5).

Why Simulations Before Training?

The risk

- Surprisal may not correlate with true turn importance on QReCC at all.
- If that correlation $\rho \approx 0$, entropy adds no signal and training compute is wasted.
- β , $\Delta t_{\min}$, τ all change faithfulness but are expensive to tune end-to-end with a 7B teacher.

The strategy

- Seven *model-free* simulations, each isolating one design question.
- Each one returns a concrete config value or a falsifiable go/no-go threshold.
- Train only if every simulation passes — otherwise the idea is rejected cheaply.

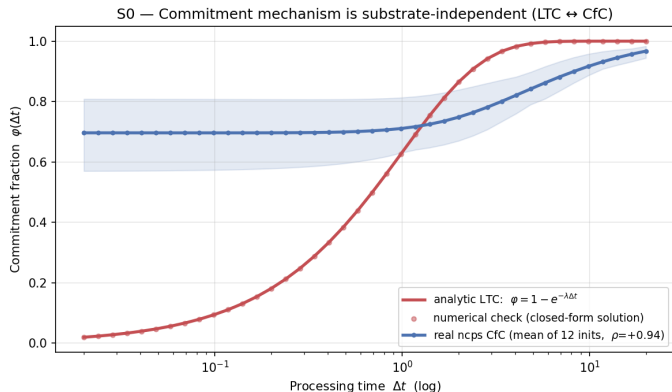
Reading the figures

Across the suite, the recurring metric is *importance recovery* — Spearman correlation between a method's per-turn score and the true latent turn importance u . Higher means the pruner keeps the turns that matter.

#	Question	Result
S0	Does the commitment lever exist in the <i>real</i> ncps CfC?	$\rho = +0.944$; analytic error $\approx 4 \times 10^{-10}$
S1	Is surprisal $\rightarrow$ commitment a tautology?	No — closed-form; but lever saturates at high S
S2	Which β^* , $\Delta t_{\min}^*$ maximise recovery?	$\beta^* = 1.0$, $\Delta t_{\min}^* = 0.05$
S5	Which Δt mapping survives outlier surprisal?	Rank normalisation most robust
S4	What does leave-one-out KL really measure?	Marginal answer-necessity = correct target
S3	Does surprisal proxy importance well enough?	Go/no-go: $\rho^* \approx 0.1$; $\rho \geq 0.5$ beats cosine
S6	What fidelity–compression does the system reach?	40% compression at 90% fidelity

The next slides walk these in order: validity (S0–S1), design (S2, S5), label (S4), the critical test (S3), and the payoff (S6).

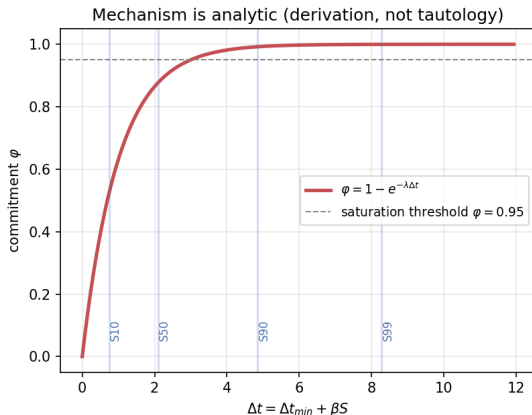
S0 — Is the Commitment Lever Real?



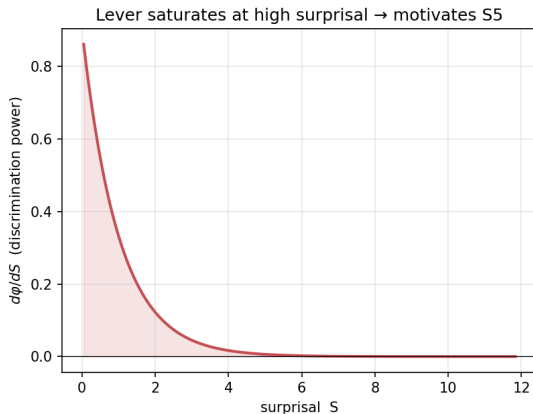
Validity foundation.

- Our analysis uses an LTC toy ($\phi = 1 - e^{-\lambda \Delta t}$); we *train* the ncps CfC.
- Both rise monotonically with Δt : real CfC tracks the analytic curve at $\rho = +0.94$.
- So a wider Δt really does “commit” the state — not an artefact of the toy model.

S1 — Derived, Not Circular — and It Saturates

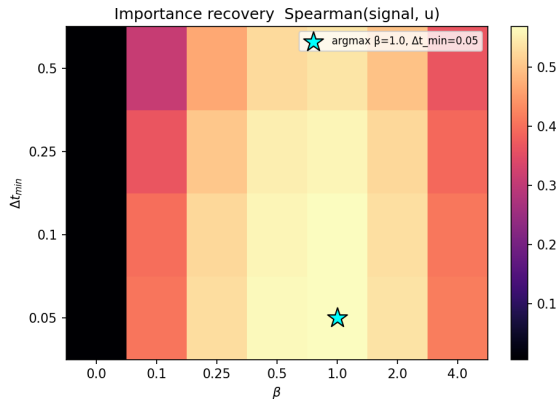


Left: commitment is a *closed-form* function of Δt — a derivation, so the surprisal $\rightarrow \varphi$ link is not a tautology.

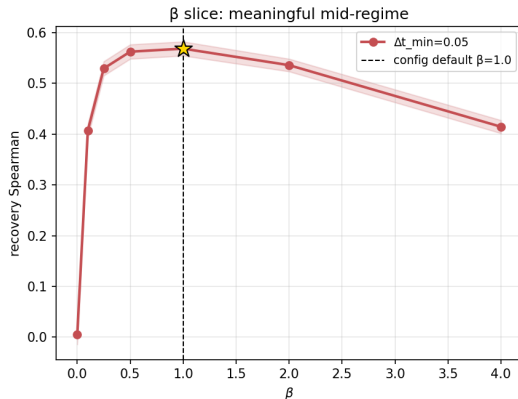


Right: $d\varphi/dS$ decays — past a point, more surprisal adds no discrimination (31.6% of turns saturate). This motivates S5.

S2 — Choosing β and $\Delta t_{\min}$ from the Data

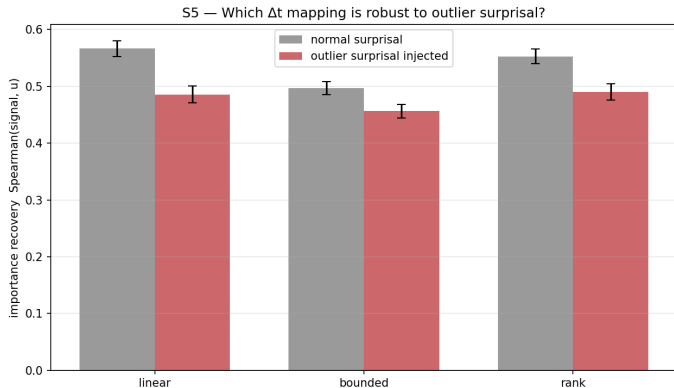


Left: recovery over the $(\beta, \Delta t_{\min})$ grid peaks at $\beta^* = 1.0, \Delta t_{\min}^* = 0.05$ (star).



Right: at $\beta = 0$ (uniform Δt) recovery ≈ 0 — the entropy term, not the network alone, carries the signal.

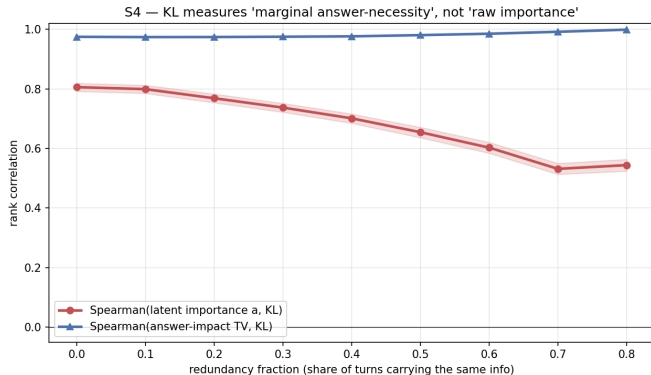
S5 — Which Δt Mapping Survives Outliers?



Why it matters (from S1).

- DistilGPT-2 surprisal is heavy-tailed: one outlier turn can monopolise a linear time budget.
- Under injected outliers, **rank** normalisation keeps the highest recovery.
- $\Rightarrow$ `build_inputs.py` maps surprisal $\rightarrow \Delta t$ by rank.

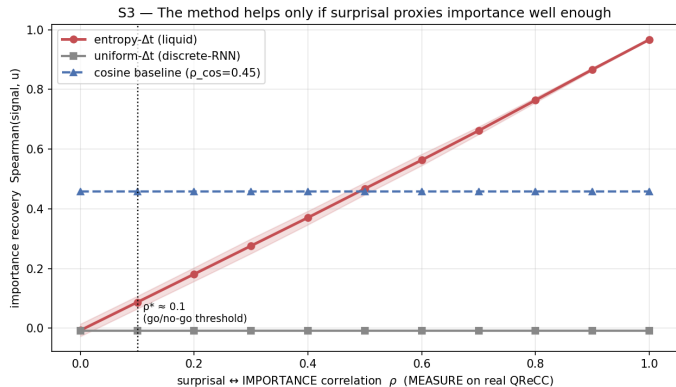
S4 — What Does the Teacher Label Measure?



Teacher labels come from **leave-one-out KL** on Mistral output.

- KL tracks answer-impact (TV) at ≈ 0.97 *regardless* of redundancy (blue).
- Its link to *raw* importance drops as turns repeat info (red, $0.81 \rightarrow 0.54$).
- So KL = *marginal* answer-necessity — exactly the right target: redundant turns are safe to prune.

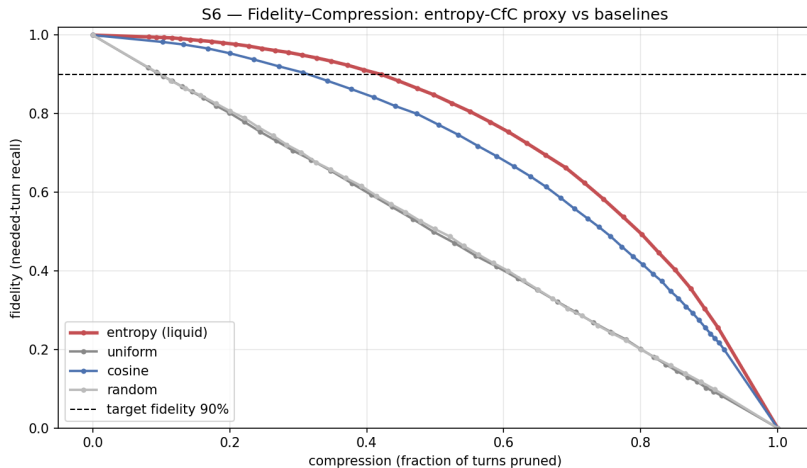
S3 — The Honest Go/No-Go



The single condition the method depends on.

- Recovery of entropy- Δt rises with $\rho(\text{surprisal, importance})$.
- Beats uniform once $\rho > \rho^* \approx 0.1$; beats the cosine baseline once $\rho \geq 0.5$.
- **Action:** measure ρ on real QReCC *before* training. Below $\rho^* \Rightarrow$ abort.

S6 — Headline: Fidelity vs. Compression



Entropy-CfC: **40% compression at 90% fidelity**. Uniform: 9%. Cosine: 31%. Random: 10%. $\Rightarrow \tau^* \approx 0.48$.

Simulation outputs $\rightarrow$ model design.

- **Input per turn:** SBERT embedding (dim 384) + rank-normalised Δt_i from DistilGPT-2 surprisal.
- **Hidden dynamics:** CfC cell; state h_i integrates over Δt_i (wider window = stronger commitment).
- **Per-turn readout:** $\hat{p}_i = \sigma(Wh_i + b)$, importance $\in [0, 1]$.
- **Teacher labels:** leave-one-out KL divergence on Mistral-7B-Instruct output distributions.
- **Loss:** MSE between $\hat{p}_i$ and normalised teacher KL.
- **Pruning:** discard turns where $\hat{p}_i < \tau^* \approx 0.48$ (from S6: 90% fidelity target).

$$\Delta t_{\min} = 0.05 \quad \beta = 1.0 \quad \tau^* \approx 0.48$$

(all three derived from simulations, not hand-tuned)

Baselines

- Full context.
- Random pruning.
- Cosine-similarity history pruning.
- Existing sentence/token LNN pruners.

Metrics

- ROUGE-L and BERTScore.
- Faithfulness with LLM-as-Judge.
- Token reduction.
- Time to first token.
- History/evidence coverage.

Current status

The CfC/LNN code and Colab notebooks are implemented as an exploratory extension, but final quantitative CfC results are not complete yet.

- Built a modular RAG pruning harness with shared pruner interface.
- Implemented no pruning, truncation, RECOMP-style extraction, LLMLingua-2, Provenance, history pruning, and combined pruning.
- Ran baseline and corrected gap-closing Colab experiments.
- Added coverage/noise metrics to diagnose pruning quality.
- Exported qualitative error-analysis candidates.
- Designed the entropy-driven CfC pruner: cognitive motivation, math derivation, and 7-simulation verification suite before training.
- Derived all hyperparameters (β^* , $\Delta t_{\min}^*$, τ^* , mapping) analytically from simulations.

- Complete human annotation and report Cohen's κ for judge calibration.
- Port the final evaluation split from HotpotQA pilot runs to QReCC.
- Tune Provenance and combined-pruning thresholds for conversational evidence coverage.
- Replace exact supporting-sentence coverage with a semantic history/evidence coverage metric.
- Finish CfC teacher-label generation and training.
- Run CfC against full context, random pruning, and cosine-history baselines.
- Evaluate on real QReCC dialogue histories instead of synthetic two-turn histories.

Main finding

Context pruning is useful, but the best pruner is not simply the most aggressive one.

- LLM-Lingua-2 gave the best corrected faithfulness score in our pilot runs.
- Combined pruning showed the strongest compression/noise reduction in pilot runs, but removed too much supporting evidence.
- CfC-style adaptive pruning is our next step for deciding *when* context should be preserved, compressed, or forgotten.

- Liu et al. 2024. Lost in the Middle: How Language Models Use Long Contexts. <https://aclanthology.org/2024.tacl-1.9/>
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